The Analysis Stage

From a research question to a result you can defend

With an example on hydropower dams and conflict

D'Amico (2026), *Global Environmental Politics*

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Today

1. What analysis is, and the main ways of doing it
2. From correlation to causation
3. Analysis: dams and conflict
4. Beyond staggered DiD: does your thesis have a "treatment"?
5. Questions, then a lab in R, in your browser

The one idea to take away

A result is the end of a chain of choices. Good analysis shows those choices, and checks whether the result changes if you make them differently.

Where we are in the course

1 Writing a thesis

Week 1

Break it into pieces and set deadlines you commit to

2 Developing an idea

Week 2

What interests you? What questions can actually be asked?

3 Literature review

Week 3

Find the puzzle: what has not been done yet, and what you add

4 Concepts and theory

Weeks 4-5

The "why": why you expect what you expect

5 Research design

Weeks 6–7

A design that matches the question

6 Conducting the research

Week 9, and today's lab

The hard part: gathering and running it

7 Analysis

Today

Interpretation, discussion, and sometimes more analyses

8 Writing the paper

Week 11

Full circle: was the question answered? Limits, surprises, and what comes next

01
What is analysis?

More than listing

> "Analysis is more than listing, more than description, more than assertion, more than a compilation of everything a researcher has done."
>
> Franklin, Chapter 7

Commit
Take a stance before you start (deduction), build one as you go (induction), or mix the two

Think
Decide what the material means and what counts as evidence

Do
Process, represent, and present it, and say what you left out

Franklin warns against **"research design by prejudice rather than consideration"**: picking quantitative or qualitative first, then forcing the question to fit.

Quantitative and qualitative analysis

Franklin puts the divide between seeing "the whole as a sum of its parts" and seeing it as "greater than the sum of its parts".

	Quantitative	Qualitative
Typical question	How much? How often? Does X change Y?	How? Why? What does it mean?
Material	Counts, surveys, events, many cases	Texts, interviews, images, few cases
Reasoning	Mostly deductive	Often inductive
Content	**Manifest**: what is said. Count it (how often "security" appears)	**Latent**: what is meant. Interpret it (how a group is portrayed)
Strength	Patterns and effects across many units	Mechanisms and context

In practice the line is blurry. A point from your comments: counting words or events can be the *first step* to interpretation, and a good quantitative paper still has to interpret.

Common ways to analyse qualitative material

Content analysis
Code texts into categories, then count or compare them. Works on *manifest* content: what is said

Thematic coding
Build themes from interviews or documents, often inductively (grounded approaches)

Discourse analysis
How language frames a problem, and who gets to define it

Textual and visual analysis
Close reading for *latent* meaning: what is meant, in context

Process tracing
Follow the steps from cause to outcome in one case. A strong way to get at the "why"

Keeping it honest
A written codebook, a second coder on a sample, and reporting where coders disagree

Content, textual, and discourse analysis are the three Franklin covers in Chapter 7.

Why I do quantitative work

- My questions are about **many places over many years**: where does conflict rise or fall after a shock?
- Data now cover the whole world at a fine scale: conflict events, dams, disasters, food prices
- Quantitative designs let me ask what would have happened **without** the shock

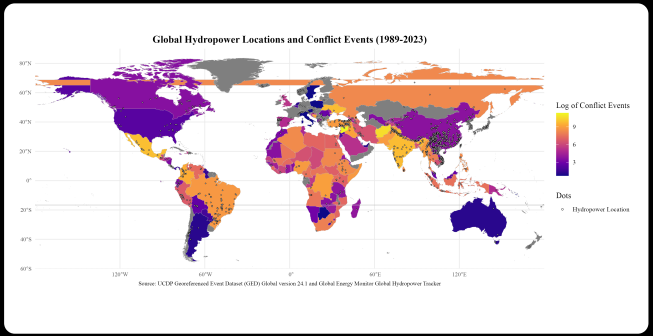
Even with statistics, Franklin says, "the researcher still needs to engage with these results by making an **interpretative intervention**."

Numbers do not speak for themselves. Choosing the unit, the comparison, and the outcome is already interpretation.

02

From correlation to causation

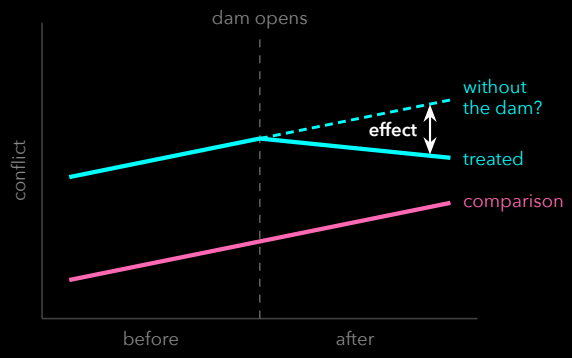
Dams and conflict on a map



- Dams and conflict often sit in the same places
- But dams are **not built at random**: rivers, terrain, money, and state reach all matter
- Those may also affect conflict
- Conflict may also stop dams being built

So the overlap on the map is a correlation. It is not yet an effect.

The question behind every causal claim



- We never see what a place would have looked like **without** its dam
- Difference-in-differences borrows that path from places that did not get a dam (yet)
- Effect = change in treated places **minus** change in comparison places

Parallel trends: without the dam, both groups would have moved together. We check it by looking at the years *before*.

Why this kind of DiD: dams arrive in different years

- The textbook version has one treatment date. Dams open in 1990, 2001, 2015…
- The old one-regression approach (two-way fixed effects) quietly uses **already-treated** places as the comparison
- Callaway and Sant'Anna (2021) compare each group only with places **not yet treated**

Country	Old way	Callaway–Sant'Anna
Pakistan	$+25.69$	$\mathbf{-19.59}$*
Sudan	$+13.18$	$+7.32$*
Colombia	-2.72	-4.81*
Philippines	-1.13*	-1.01*

Effect on conflict events per grid cell-year. * 95% interval excludes zero. D'Amico (2026), Table A.I.

> In Pakistan the sign flips. The comparison you choose can change the answer.

03

Analysis: dams and conflict

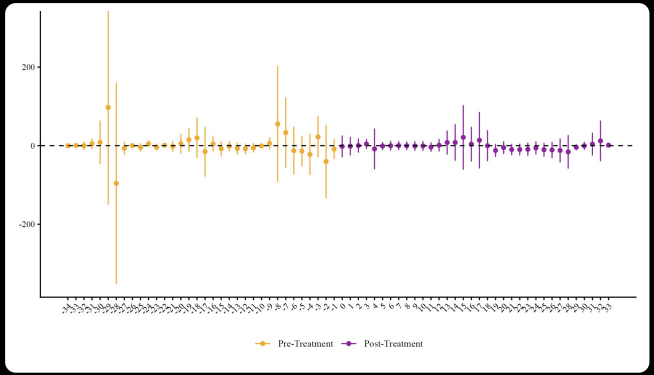
The design, one choice at a time

Choice	What the paper does	Could have been…
Unit	Hexagonal grid cells, 1.25° (about 140 km across)	Districts, provinces, finer cells
Treatment	First dam of 75 MW or more (Global Hydropower Tracker)	Any dam, capacity, dam status
Outcome	Armed conflict events (UCDP)	Deaths, civilian deaths
Comparison	Cells not yet treated	Cells never treated
Timing	Effects may start a year early (construction)	No anticipation
Cases	8 countries, 1989–2024	Pooled across all countries

Franklin calls this "making a commitment". Every row is one.

Why hexagons? (from your comments) Same shape and size everywhere, and no borders drawn for political reasons. 1.25° is "a balance between granularity and coherence": smaller cells add zeros and noise, bigger ones blur places together.

Pooled across all eight countries: nothing

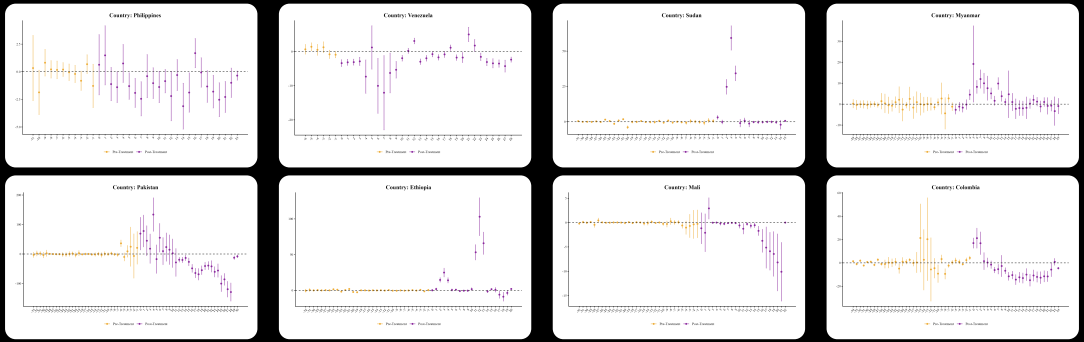


- **x** years before and after the first dam
- **y** effect on conflict events
- **Orange** before, **purple** after
- Estimates bounce around zero

> Stop here and the conclusion is "dams have no effect on conflict".

D'Amico (2026), *Global Environmental Politics*, Figure 2.

Country by country: three different stories



Falls Philippines, Venezuela **Rises, then returns** Sudan, Myanmar **Disruption, then falls** Pakistan, Ethiopia, Mali, Colombia

So what can a policymaker learn?

From your comments: if the pooled result is zero and countries differ, what is the general lesson?

- The pooled zero hides real effects that cancel out
- "Dams cause conflict" and "dams bring peace" are both too simple
- The useful question becomes: **where** and **under what conditions**?

DiD tells you *whether* and *when* conflict changed. It cannot tell you *why*.

Governance, who gets displaced, who gets the electricity and the jobs: that is where case studies and process tracing come in.

The data do not speak for themselves (Franklin). Back to quantitative *and* qualitative.

Robustness: does the result hold up?

Country	Choice changed	Before	Alternative	After	Verdict
Colombia	Comparison group	-4.81*	never treated	-4.80*	Robust
Venezuela	Comparison group	-2.51*	never treated	-2.51*	Robust
Colombia	Anticipation	-4.81*	none	-7.65*	Same sign, bigger
Venezuela	Anticipation	-2.51*	none	-1.60*	Same sign, smaller
Philippines (disasters)	Anticipation	$+2.37$*	one year	$+1.05$	Fragile

Effect on conflict events per grid cell-year. * 95% interval excludes zero. My reruns of the paper's model.

Robust: same sign, similar size. **Fragile:** the sign flips, the size jumps, or the interval now includes zero.

Change **one** thing at a time, and only to choices you could defend. You do this in Step 4 of the lab.

What I'd look at more carefully now

Country	With a dam	Used
Philippines	3	**1**
Venezuela	4	**1**
Colombia	14	7
Mali	2	2
Sudan	3	3
Ethiopia	8	7
Pakistan	11	10
Myanmar	22	21

- Dams that opened before 1991 have no "before" period, so the method drops those cells
- The Philippines and Venezuela results each rest on **one** grid cell
- Some dams are only planned or under construction. All start 2022–24, so they barely matter here, but I would flag it

> From your comments: it is easy to treat a significant coefficient as the conclusion. This is why: the asterisk says nothing about how many places stand behind it.

Same design, different outcome

Country	Conflict events	Total deaths
Sudan	$+7.32$*	$+76.14$*
Myanmar	$+2.41$*	$+26.20$
Pakistan	-19.59*	-248.68*
Ethiopia	$+12.57$*	-41.64

Effect per grid cell-year, paper's specification. * 95% interval excludes zero.

- "Conflict" can mean how often fighting happens or how deadly it is
- In Ethiopia the sign flips, with a huge margin of error: a few war years drive the deaths
- Neither outcome is *the* right one. Which you report, and why, is part of the analysis

04
Beyond staggered DiD

When everyone is treated at once

A thesis example. When the US left the Paris Agreement, did the small-island coalition speak up more in UN climate talks?

% of speaking turns	US in	US out	Change
United States	5.17	2.12	−3.05
Small islands	4.09	4.71	+0.62
EU (comparison)	7.09	7.54	+0.45
DiD: islands' change minus EU's change			≈ 0

Averages over 19 negotiating meetings after Paris (11 with the US in, 8 during its withdrawal). DiD estimate $+0.18$ points, SE 0.92. A quick illustration, not the thesis model.

- **First, check the treatment happened.** The US spoke far less
- **The answer:** the islands did not speak up more than the EU did
- **On loss and damage only,** the islands' share jumps from about 4% to 18%. Promising, but with 19 meetings it could be chance

Everyone is hit at once, so no group is "not yet treated". The comparison is a group that is **less exposed**: a classic 2×2 DiD.

Which kind of DiD fits your question?

Your treatment…	Example	Design
hits different units at **different times**	Dams, minimum wage rises	Staggered DiD (Callaway–Sant'Anna)
hits **everyone at once**	US leaves Paris, Brexit vote	2×2 DiD on more vs less exposed units
switches **on and off**	A policy reversed, then restored	Possible, but harder
does not exist	"Is civil disobedience justified?"	Not DiD, and that is fine

Start from the question, then pick the design. Doing it the other way round is Franklin's "research design by prejudice". Step 6 of the lab asks you this about your own thesis.

Discussion: could disasters be the treatment?

Same grid, same method. The treatment is now a cell's first flood, storm, or other climate disaster (EM-DAT).

- No one *chooses* where a flood hits. Is that better than dams?
- EM-DAT gives map coordinates for only about one event in four, and only for 2001–2020
- Philippines: $+2.37$* with no anticipation, $+1.05$ (not distinguishable from zero) with one year

Talk to your neighbour

1. Some "never treated" cells were probably hit too. What does that do to the comparison?
2. Floods hit the same places again and again. Is the first one the right treatment?
3. How would you measure the treatment better?

You can try it yourself in Challenge E.

Questions

What would you like to ask?

From your comments on the readings:

- How confidently can qualitative work explain the "why"?
- How do you keep bias out of a coding scheme?
- Is Franklin too harsh on counting?
- How do you pick the unit of analysis?

05
Lab

The lab, step by step

	Step	You do	Look for
0	Look at the data	Open the panel and the dam list	Cells that appear twice
1	Build the treatment	With dplyr: each cell's first dam, merged onto the panel	Countries with very few treated cells. Which dams count?
2	Reproduce the paper	Run the paper's model for the Philippines	-1.007? Flat before year 0?
3	Explore	`explore()` 3+ countries, outcomes, and disasters	Usable cells, spread of dam years, parallel trends
4	Your analysis	Pick a country, then change one choice at a time	Robust or fragile?
5	Challenge	Pick one of A–E. No code given	One honest sentence for a paper
6	Your thesis	On your own: is there a treatment? Which DiD fits?	What would change your mind

The lab: R in your browser, nothing to install



www.elisadamico.net/analysis-stage-lab

Code and data: github.com/elisadamico/analysis-stage-lab

- Run it **in order from the top**. R only knows what you have already run
- Steps 0–5 **in pairs**: one person who has coded before with one who has not. **The one who has not types**
- Step 6 **on your own**
- Run a line with `Ctrl`/`Cmd` + `Enter`. The guide at the top of the page explains every button
- Write answers after `# ANSWER:`
- **Download your script** at the end
- Then click Give feedback. It is anonymous